

7. Provide a brief report explaining:

1. Which imputation strategy was most effective?

The **MICE (Multiple Imputation by Chained Equations)** strategy was the most effective because it estimates missing values using relationships between multiple variables. It preserves the overall data pattern better than simple methods such as mean, median, or mode imputation.

2. Which outlier handling method preserved data quality best?

The **IQR Capping (Winsorization)** method preserved data quality best. Instead of removing records, it replaced extreme values with the lower and upper IQR limits, reducing the effect of outliers while keeping all observations.

3. How did data cleaning improve dataset usability?

- Missing values were handled, making the dataset complete.
- Outliers were treated, reducing the influence of extreme values.
- Data quality and consistency improved.
- The cleaned dataset became more reliable for visualization, statistical analysis, and machine learning.